# Model Report – Credit Scoring for Home Equity Loans

## 1. Executive Summary

The objective of this project was to develop a transparent and robust credit scoring model for home equity loans, based on borrower characteristics and credit behavior. The dataset consists of 5,960 observations, including information on loan characteristics, credit history, and delinquency indicators, with loan default defined as the target event.

The development followed standard credit risk modeling practice. First, an extensive univariate analysis was conducted to assess the explanatory power of individual variables. Continuous variables were discretized into approximately equally sized bins, while categorical variables were analyzed using their natural groupings. Default rates, Weight of Evidence (WoE), and Information Value (IV) were used to evaluate discriminatory power and monotonicity with respect to credit risk.

Based on this analysis, a shortlist of variables with sufficient discriminatory power and economic interpretability was selected. Variables exhibiting low information value or strong mutual correlations were excluded to avoid instability in the subsequent multivariate model. Particular attention was paid to variables with a high concentration of zero or missing values, where binning strategies were adjusted to better reflect their economic meaning.

A logistic regression model was then developed using WoE-transformed variables. Model performance was evaluated using repeated random splits of the dataset into training and validation samples. The model demonstrated strong and stable discriminatory power, with Gini coefficients consistently around 76–77% for both samples, indicating good generalization and no signs of overfitting.

For operational usability, the final logistic regression model was transformed into a scorecard using standard banking conventions. The resulting score provides an intuitive and interpretable measure of credit risk, where higher scores correspond to lower probability of default. Decile analysis confirms strong ranking ability, with a clear overall increasing trend in observed default rates across score deciles.

Overall, the developed scorecard represents a robust, interpretable, and practically applicable credit scoring solution suitable for use in a banking environment.

## 1. Data Description

The dataset contains information on **5,960 home equity loans**, including borrower characteristics and credit history. The dependent variable is:

- **BAD**:
    - 1 = applicant defaulted or became seriously delinquent

- 0 = loan paid as agreed

The explanatory variables include:

- **LOAN –** Loan amount
- **MORTDUE –** Outstanding mortgage balance
- **VALUE –** Property value
- **REASON –** Loan purpose (DebtCon, HomeImp)
- **JOB** – Occupational category
- **YOJ** – Years at current job
- **DEROG** – Number of major derogatory reports
- **DELINQ –** Number of delinquent credit lines
- **CLAGE –** Age of oldest credit line (months)
- **NINQ –** Number of recent credit inquiries
- **CLNO –** Number of credit lines
- **DEBTINC –** Debt-to-income ratio

All calculations were performed using **R statistical software**.

## 2. Univariate Analysis and Variable Preselection

In the univariate analysis, continuous variables were divided into approximately five equally sized bins, while categorical variables were grouped into natural categories. For each bin, the **default rate** (share of observations with BAD = 1) was calculated.

Variables were evaluated based on:

- their ability to discriminate between defaulted and non-defaulted clients, and
- the monotonicity of the relationship between the variable and credit risk.

### Key findings:

- **CLAGE** shows a clear monotonic decline in default rate with increasing values, confirming the expected negative relationship between credit history length and default probability.
- **DEROG, DELINQ, and NINQ** exhibit a strong concentration of zero values. Standard quantile binning was not interpretable; therefore, these variables were rebinned to explicitly distinguish between zero and non-zero values, reflecting their economic meaning and improving model stability.
- Categories of **JOB** were merged into three groups due to similar default rates.
- **REASON** and **CLNO** were excluded due to nearly constant default rates across categories.

- **DEBTINC** shows exceptionally strong discriminatory power. A substantial part of this power comes from missing values, which is consistent with credit scoring practice where missing information often signals increased risk. DEBTINC contains 1,267 missing observations, of which 62% correspond to defaults.
- **LOAN, MORTDUE, VALUE, and YOJ** show a weak and only approximate monotonic relationship with default risk.

Selected default rate summaries are shown in **Table 1**.

*Table 1: Analysis of default rates.*

| CLAGE bin | defaults_no/bin_rows | bin min_value | bin max_value | YOJ bin | defaults_no/bin_rows | bin min_value | bin max_value |
|---|---|---|---|---|---|---|---|
| 1 | 0.3017699 | 0 | 105.7496 | 1 | 0.2359963 | 0 | 2 |
| 2 | 0.2513274 | 105.768 | 145.1 | 2 | 0.2460973 | 2 | 5 |
| 3 | 0.1840708 | 145.1 | 193.0739 | 3 | 0.1735537 | 5 | 9 |
| 4 | 0.1433628 | 193.0815 | 246.9909 | 4 | 0.2130395 | 9 | 15 |
| 5 | 0.1024735 | 247.0373 | 1168.234 | 5 | 0.1634527 | 15 | 41 |

| DEROG bin | defaults_no/bin_rows | bin min_value | bin max_value |
|---|---|---|---|
| 1 | 0.1665562 | 0 | 0 |
| 2 | 0.48 | 1 | 10 |

| JOB bin | defaults_no/bin_rows | cathegory | REASON bin | defaults_no/bin_rows | cathegory |
|---|---|---|---|---|---|
| 1 | 0.3178808 | "Sales" + "Self" | 1 | 0.2224719 | "HomeImp" |
| 2 | 0.2323296 | "Mgr" + "Other" | 2 | 0.189664 | "DebtCon" |
| 3 | 0.1515288 | "ProfExe" + "Office" | | | |

## 3. Weight of Evidence and Information Value

To quantify discriminatory power more formally, **Weight of Evidence (WoE)** and **Information Value (IV)** were calculated.

For each bin j, WoE is defined as:

$$\mathrm{WoE}_j = \ln\left(\frac{G_j / G}{D_j / D}\right),$$

where $G_j$ and $D_j$ denote the number of non-defaults and defaults in bin j, and G and D are their totals in the dataset.

The **Information Value** is defined as:

$$IV = \sum_j \left( \frac{G_j}{G} - \frac{D_j}{D} \right) \cdot WoE_j \ .$$

IV values above **0.1** are considered indicative of meaningful predictive power.

### Results (Table 2):

The variables **CLAGE, DEBTINC, DELINQ, DEROG, and NINQ** show sufficient discrimination power and were selected for model development. Other variables were excluded due to low IV values and potential instability in multivariate modeling.

Table 3 illustrates the monotonic WoE pattern for CLAGE, which is typical for variables with strong predictive power, while YOJ shows a weaker and less stable pattern.

*Table 2: Information Value of variables.*

| CLAGE | DEBTINC | DELINQ | DEROG | JOB |
|---|---|---|---|---|
| 0.2155175 | 1.778838 | 0.5135998 | 0.3442104 | 0.08384423 |
| LOAN | MORTDUE | NINQ | VALUE | YOJ |
| 0.0878338 | 0.04899977 | 0.1232794 | 0.05248935 | 0.0411278 |

*Table 3: WoE and IV of bins of variables CLAGE and YOJ.*

| CLAGE bin | default | non-default | Woe | iv_bin | YOJ bin | default | non-default | woe | iv_bin |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 789 | 341 | -0.5690029 | 0.075780044 | 1 | 832 | 257 | -0.17183675 | 0.0062032492 |
| 2 | 846 | 284 | -0.3163416 | 0.021929746 | 2 | 821 | 268 | -0.22705698 | 0.0109967893 |
| 3 | 922 | 208 | 0.0811204 | 0.001283346 | 3 | 900 | 189 | 0.21405464 | 0.0085912229 |
| 4 | 968 | 162 | 0.3797490 | 0.025577738 | 4 | 857 | 232 | -0.03989256 | 0.0003220117 |
| 5 | 1016 | 116 | 0.7621517 | 0.090946665 | 5 | 911 | 178 | 0.28616624 | 0.0150145237 |

## 4. Model Development

The selected variables (**CLAGE, DEBTINC, DELINQ, DEROG, NINQ**) were transformed into their corresponding WoE values based on the assigned bins. These WoE-transformed variables were then used as inputs to a **logistic regression model**.

The model estimates the log-odds of default as:

$$\ln\left(\frac{p}{1-p}\right) =$$

$$= \beta_0 + \beta_1 \cdot CLAGE\_WOE + \beta_2 \cdot DEBTINC\_WOE + \beta_3 \cdot DELINQ\_WOE + \beta_4 \cdot DEROG\_WOE + \beta_5 \cdot NINQ\_WOE$$

Estimated coefficients, standard errors, and z-statistics are shown in **Table 4**. All variables are statistically significant and exhibit economically meaningful signs.

*Table 4: Coefficients of logistic regression.*

| Coefficient | Estimate | Std. Error | z value | Pr(>\|z\|) |
|---|---|---|---|---|
| $\beta_0$ | -1.28120 | 0.04652 | -27.540 | <2e-16 |
| $\beta_1$ | -1.08571 | 0.10181 | -10.664 | <2e-16 |
| $\beta_2$ | -0.92864 | 0.03218 | -28.855 | <2e-16 |
| $\beta_3$ | -0.92314 | 0.06236 | -14.804 | <2e-16 |
| $\beta_4$ | -0.73867 | 0.07656 | -9.648 | <2e-16 |
| $\beta_5$ | -0.33330 | 0.13023 | -2.559 | 0.0105 |

## 5. Model Performance and Validation

To assess model stability and avoid biased performance estimates, the dataset was randomly split into **training (70%)** and **validation (30%)** samples across five independent runs.

Model performance was evaluated using:

- **AUC (Area Under the ROC Curve)**
- **Gini coefficient**, defined as

$$\text{Gini} = 2 \cdot \text{AUC} - 1.$$

AUC represents the probability that a randomly selected default observation receives a higher risk score than a randomly selected non-default observation.

### Results:

The Gini coefficients for training and validation samples are shown in **Table 5**. The results are stable across runs, with very similar in-sample and out-of-sample values, indicating good generalization and no evidence of overfitting.

*Table 5: Full Model GINI.*

| Run | training sample | validation sample |
|---|---|---|
| 1 | 76.86% | 76.04% |
| 2 | 77.20% | 75.37% |
| 3 | 76.01% | 77.90% |
| 4 | 76.02% | 77.70% |
| 5 | 76.82% | 75.90% |

## 6. Final Model and Decile Analysis

Based on the final run, a reference model was selected. Its coefficients (Table 6) are consistent with those estimated on the full dataset.

A **decile analysis** was performed on the validation sample. Observations were sorted by predicted default probability and divided into ten equally sized groups.

As shown in **Table 7**, the observed default rate increases near-monotonically from the lowest to the highest risk decile, confirming strong ranking ability and practical usability of the model.

*Table 6: Coefficients of logistic regression for training sample.*

| Coefficient | Estimate | Std. Error | z value | Pr(>\|z\|) |
|---|---|---|---|---|
| $\beta_0$ | -1.26660 | 0.05540 | -22.862 | <2,00E-16 |
| $\beta_1$ | -0.99416 | 0.12110 | -8.209 | 2.23e-16 |
| $\beta_2$ | -0.92110 | 0.03840 | -23.988 | <2,00E-16 |
| $\beta_3$ | -0.89554 | 0.07381 | -12.132 | <2,00E-16 |
| $\beta_4$ | -0.83099 | 0.09096 | -9.136 | <2,00E-16 |
| $\beta_5$ | -0.39855 | 0.15581 | -2.558 | 0.0105 |

*Table 7: Decile analysis.*

| Decile | n | bad_rate |
|---|---|---|
| 1 | 162 | 0.02469136 |
| 2 | 141 | 0.02127660 |
| 3 | 168 | 0.01785714 |
| 4 | 134 | 0.04477612 |
| 5 | 166 | 0.09036145 |
| 6 | 144 | 0.07638889 |
| 7 | 143 | 0.20979021 |
| 8 | 151 | 0.17218543 |
| 9 | 160 | 0.51250000 |
| 10 | 143 | 0.84615385 |

For practical implementation in a banking environment, the final logistic regression model was subsequently transformed into a scorecard using standard scaling conventions.

# 7. Scorecard Scaling

## Scorecard Scaling and Interpretation

To enable practical implementation and interpretation in a banking environment, the final logistic regression model was transformed into a scorecard. The transformation maps the predicted probability of default (PD) into a numerical score, where higher values indicate lower credit risk. Let p denote the predicted probability of default from the logistic regression model. The score is defined as:

$$\text{Score} = \text{Offset} - \text{Factor} \cdot \ln\left(\frac{p}{1 \quad p}\right).$$

This formulation ensures a linear relationship between the score and the log-odds of default.

### Scaling Parameters

Standard industry conventions were applied for score scaling:

- **Points to Double the Odds (PDO):** 20
- **Reference score:** 600 points
- **Reference odds:** 50:1 (non-default to default); *since the score formula is based on the default odds* $p/(1-p)$, *the reference odds of 50:1 (non-default to default) correspond to default odds of 1:50*

Based on these assumptions, the scaling parameters are given by:

$$\text{Factor} = \frac{\text{PDO}}{\ln 2}, \qquad \text{Offset} = 600 - \text{Factor} \cdot \ln 50.$$

This calibration ensures that an increase of 20 score points corresponds to a doubling of the odds of non-default (equivalently, a halving of the odds of default).

### Interpretation

Under this scorecard:

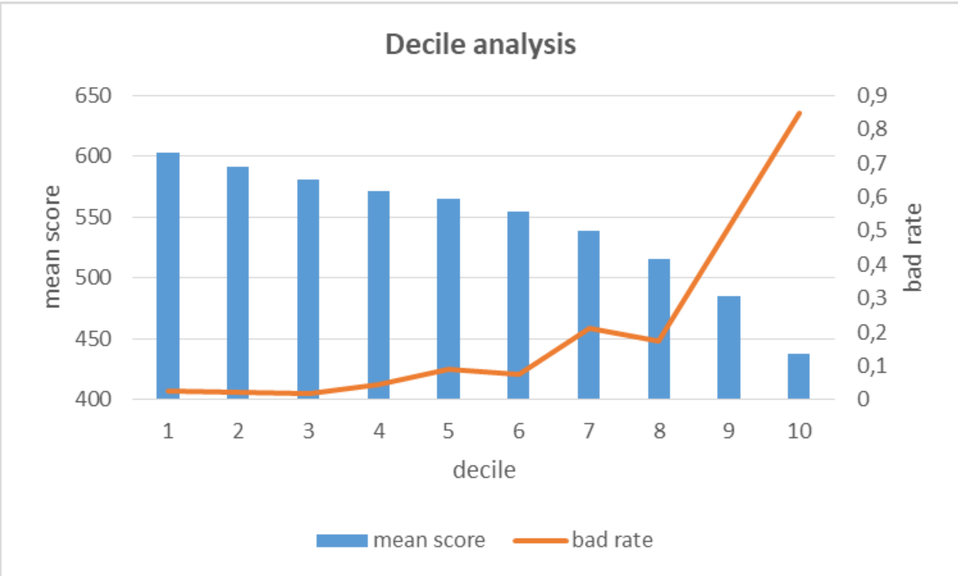
- Higher scores indicate lower credit risk.
- Each increase of 20 points halves the odds of default.
- The score provides an intuitive ranking of applicants while preserving the discriminatory power of the underlying logistic regression model.

In Table 8, the scorecard was validated using a decile analysis on the validation sample. As shown in Table 7, the observed default rates demonstrate near-monotonic behavior across risk deciles, providing evidence that the scorecard effectively ranks applicants according to credit risk. This demonstrate the plots in Fig. 1 as well.

*Table 8: Scorecard of deciles on validation sample.*

| decile | n | bad_rate | mean_score |
|---|---|---|---|
| 1 | 162 | 0.02469136 | 602.7445 |
| 2 | 141 | 0.02127660 | 591.2399 |
| 3 | 168 | 0.01785714 | 581.2925 |
| 4 | 134 | 0.04477612 | 571.6488 |
| 5 | 166 | 0.09036145 | 564.5233 |
| 6 | 144 | 0.07638889 | 554.7128 |
| 7 | 143 | 0.20979021 | 538.9750 |
| 8 | 151 | 0.17218543 | 515.1737 |
| 9 | 160 | 0.51250000 | 484.5770 |
| 10 | 143 | 0.84615385 | 438.1672 |

*Figure 1: Scorecard and bad rate of deciles on validation sample.*




## 8. Conclusion and Recommendations

The developed scoring model demonstrates strong discriminatory power, stability across samples, and clear interpretability. The use of WoE transformation ensures robustness and alignment with industry-standard credit scoring practices.

The model is suitable for use as a baseline application scorecard. Future enhancements may include:

- testing alternative modeling approaches (e.g. tree-based or regularized models),
- incorporating reject inference techniques, or
- periodic recalibration on newer data.

Reported values may slightly differ from the accompanying refactored scripts due to minor implementation differences in bin assignment and category grouping. The overall conclusions, variable selection, and model performance remain unchanged.